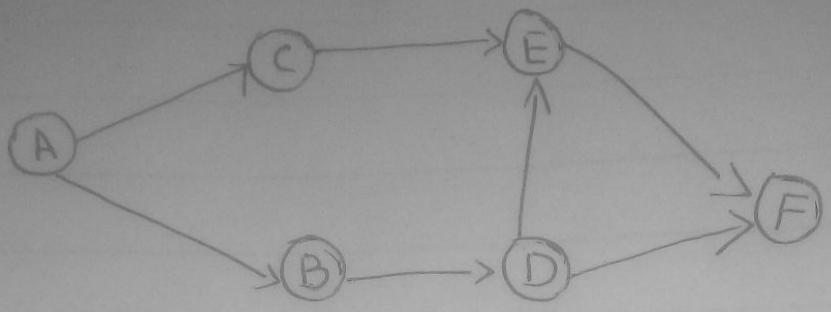
Q:- Give Flow Diagram and Find the Predecessor.



Solution:-

Event	Predecessor
A	-
B	A
C	A
D	B, [redacted]
E	D, C
F	D, E

Exam main aise Diagram de kr Predecessor Puchte hain -

11-12-2023 - 12:30

Question:- MTH601 BY Mathematics

Find the Total Float and Independent Float. Table given.

Activity	Duration	EST	EFT	LFT	LST
1-2	3	2	3	3	0
2-3	0	1	2	0	1
3-4	2	5	1	4	2

Solution:-

Total Float

$T = LFT - EST - D$

Independent Float:-

$I = EFT - LST - D$

Now we find Total & Independent Float.

Activity	Duration	EST	EFT	LFT	LST	Total Float LFT-EST-D	Independent EFT-LST-D
1-2	3	2	3	3	0	3-2-3=2	3-0-3=0
2-3	0	1	2	0	1	0-1-0=-1	2-1-0=1
3-4	2	5	1	4	2	4-5-2=3	1-2-2=-4

یہ والیو minus میں نہیں آنی ہوتی کبھی نہیں آنی یاد

اس لیے میں نے اپنے سے والیو لی ہیں اسی

اس سے ہو گا سوال minus کو ignore کریں method

Just Concept Clear kren

Same aise Question tha

Question:-

$C_2 = 1$ Demand = 200, $C_3 = 25$

EOQ = ? Find the economic order Quantity.

Solution:-

EOQ Formula

$$Q^* = \sqrt{\frac{2C_2D}{C_3}}$$

$$Q^* = \sqrt{\frac{2(1)(200)}{25}}$$

$$Q^* = \sqrt{\frac{400}{25}}$$

$$Q^* = \sqrt{16}$$

$$Q^* = 4$$

Time of order =

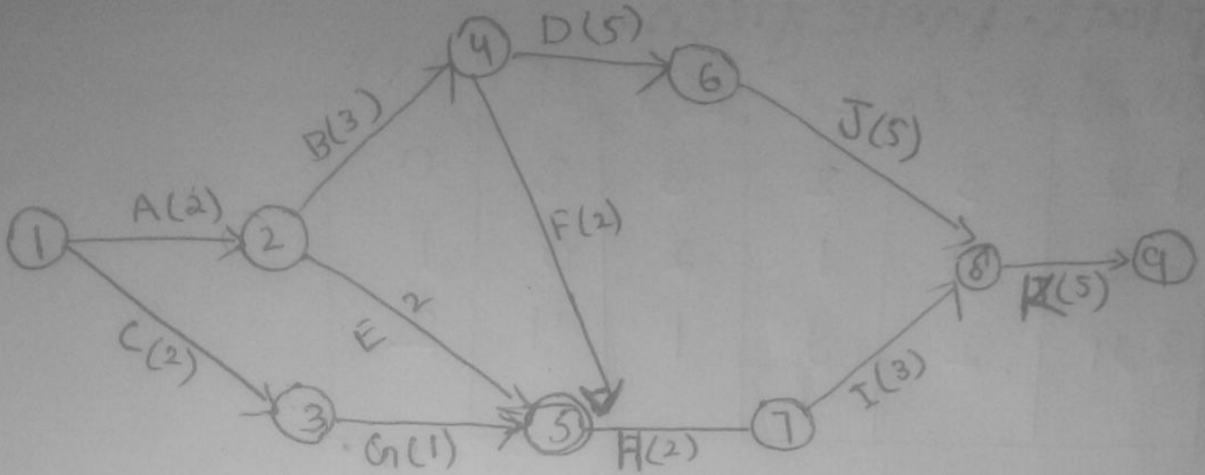
$$t = \frac{Q^*}{D} = \frac{4}{200} = 0.02$$

No. of order

$$N = \frac{D}{Q^*} = \frac{200}{4} = 50.$$ Answer.

Question:-

Find the Critical Path and Completion time.



Solution:-

Critical Path

① 1-2-4-6-8-9 = A-B-D-J-K = 2+3+5+5+5 = 20

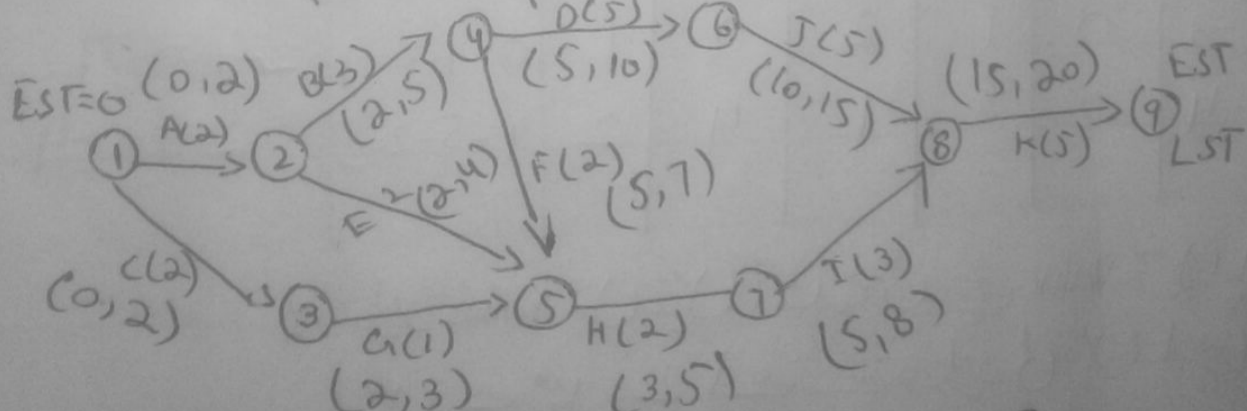
② 1-2-5-7-8-9 = A-E-H-I-K = 2+2+2+3+5 = 14

③ 1-2-4-5-7-8-9 = A-B-F-H-I-K = 2+3+2+2+3+5 = 17

④ 1-3-5-7-8-9 = C-G-H-I-K = 2+1+2+3+5 = 13

So, the Maximum Critical Path is = 20

and we find Completion Time in Diagram



So, Completion Time is 20

4 Manfacturing Model with Shortage

$$Q^* = \sqrt{\frac{2C_2D}{C_3(1-D/R)}}\sqrt{\frac{C_3+C_4}{C_4}}$$

and

$$S^* = \sqrt{\frac{2C_2D}{C_4}}\sqrt{1-D/R}\sqrt{\frac{C_3}{C_3+4}}$$

5 Expected Time:

$$Te = \frac{t_o + 4tm + t_p}{6}$$

6 Standard deviation:

$$S.D = \frac{t_p - t_o}{6}$$

7 Variance:

$$V = \left(\frac{t_p - t_o}{6}\right)^2$$

8 Expected Project length:-

$$Z = \left[\frac{T - Te}{\sigma}\right]$$

1 Purchasing Model with no Shortage
(willson method) ∴ $C_4 = 0$

$$Q^* = \sqrt{\frac{2C_2D}{C_3}} \qquad t^* = \frac{Q^*}{D}$$

No of order :-

$$N^* = \frac{D}{Q^*}$$

2 Purchasing model with Shortage

$$Q^* = \sqrt{\frac{2C_2D}{C_3}} \sqrt{\frac{C_3+C_4}{C_4}}$$

$$S^* = \sqrt{\frac{2C_2D}{C_4}} \sqrt{\frac{C_3}{C_3+C_4}}$$

3 (Manufacturing Model with no Shortage)

$$Q^* = \sqrt{\frac{2C_2D}{C_3}} \sqrt{\frac{R}{R-D}}$$

Time b/w order :-

$$t = \frac{Q}{D}$$

Num of order :

$$N = D/Q$$

Formulas

1. Early Start and Early Finish Time
   EST and EFT

   EFT = EST + Duration of the activity

2. Latest Start and Latest Finish Time
   LST and LFT

   LST = LFT − Duration of the activity

3. Float or Slack (same)
   1. Independent Float/Slack
   2. Free Float/Slack
   3. Total Float/Slack

1. Independent Float/slack:−

   I = EFT − LST − D

Free Float/slack:−

   F = EFT − EST − D

Total Float/slack:−

   T = LFT − EST − D

12:00 11-12-2023 MTH601 BY Mathematics

Current Paper Subjective

Question:1 3 Marks

Given:

Items no	Annual usage	Rank
1	3000	6
2	40500	1
3	300	9
4	5000	4
5	200	10
6	22000	2
7	800	7
8	4000	5
9	9000	3
10	7000	8

Calculate the Comulative usage and % Comulative.

Solution:-

			Annual usag	Annual usage of item
1	2	40500	40500	
2	6	22000	40500+22000= 62500	
3	9	9000	62500+9000=71500	
4	4	5000	71500+5000=76500	
5	8	4000	same 80500	
6	1	3000	83500	
7	7	800	84300	
8	10	700	85000	
9	3	300	85300	
10	5	200	85500	
			Total 755300	